Parametric Differentiation

Exam code: 9MA0

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Help me understand

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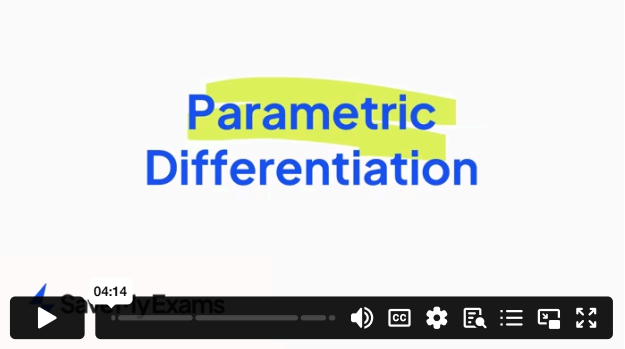

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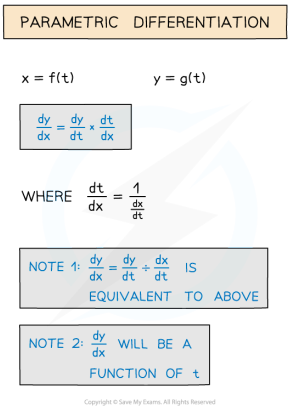
Parametric differentiation

How do I find dy/dx from parametric equations?

- Ensure you are familiar with Parametric Equations – Basics first



- This method uses the chain rule and the reciprocal property of derivatives
  - $\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx}$

Stuck on something?

Ask questions and get help with exactly what you're revising.

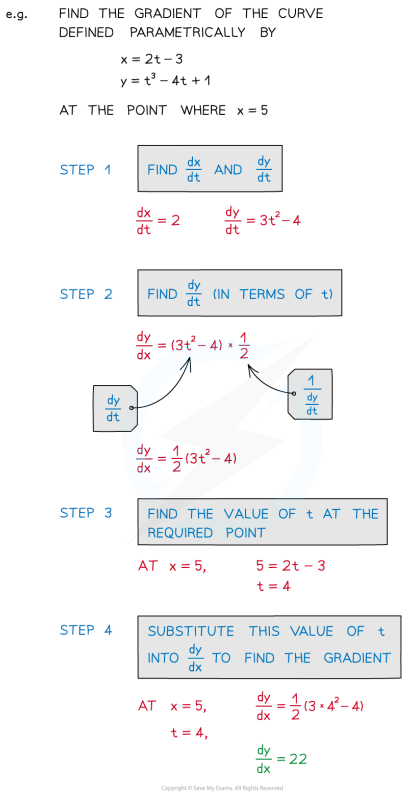
Got it

- Equivalently, $\frac{dx}{dt} \cdot \frac{dt}{dt} = \frac{dt}{dt}$
- $\frac{dy}{dx}$ will be in terms of t – this is fine
  - Questions usually involve finding gradients, tangents and normals
- The chain rule is always needed when there are three variables or more – see Connected Rates of Change

How do I find gradients, tangents and normals from parametric equations?

To find a gradient

- **STEP 1**
  Find **dx/dt** and **dy/dt**
- **STEP 2**
  Find **dy/dx** in terms of **t**
  - Using either **dy/dx = dy/dt ÷ dx/dt**
    or **dy/dx = dy/dt × dt/dx** where **dt/dx = 1 ÷ dx/dt**
- **STEP 3**
  Find the value of **t** at the required point
- **STEP 4**
  Substitute this value of **t** into **dy/dx** to find the gradient



To then go on to find the equation of a tangent

- **STEP 5**
  Find the **y** coordinate

e.g. FIND THE EQUATION OF THE TANGENT TO THE CURVE DEFINED PARAMETRICALLY BY

$$x = 2t - 3$$
$$y = t^3 - 4t + 1$$

AT THE POINT WHERE $x = 5$

TANGENT: "$y - y_1 = m(x - x_1)$"

$m = \frac{dy}{dx}$ AT $x = 5$

FROM STEP 4 → $m = 22$

STEP 5 FIND THE y-COORDINATE

TO FIND THE EQUATION OF A LINE WE NEED THE GRADIENT, m, AND A POINT (x_1, y_1) THE LINE PASSES THROUGH

AT $x = 5$, $t = 4$, $y = 4^3 - 4 \cdot 4 + 1 = 49$

FROM STEP 3

STEP 6 USE GRADIENT AND POINT TO FIND EQUATION OF TANGENT

$y - 49 = 22(x - 5)$

$y = 22x - 61$

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To find a normal...

- **STEP 7**
  Use perpendicular lines property to find the gradient of the normal $\mathbf{m_1 \times m_2 = -1}$
- **STEP 8**
  Use gradient and point to find the equation of the normal $\mathbf{y - y_1 = m(x - x_1)}$

How do I solve problems using parametric differentiation?

- Questions may require use of tangents and normals as per the coordinate geometry sections
  - Find points of intersection between a tangent/normal and **x/y** axes
  - Find areas of basic shapes enclosed by axes and/or tangents/normals

- Find stationary points (**dy/dx = 0**)

$y = t^3 - 4t + 1$

$\frac{dy}{dx} = \frac{1}{2}(3t^2 - 4)$ ← FROM EARLIER EXAMPLES

$\frac{1}{2}(3t^2 - 4) = 0$ ← $\frac{dy}{dx} = 0$ AT STARTING POINTS

$t^2 = \frac{4}{3}$

$t = \pm\frac{2}{\sqrt{3}}$ → SAME AS $t = \pm\frac{2\sqrt{3}}{3}$

WORK OUT x & y FROM t

AT $t = \frac{2}{\sqrt{3}}$, $x = \frac{-9 + 4\sqrt{3}}{3}$ $y = \frac{9 - 16\sqrt{3}}{9}$

AT $t = \frac{-2}{\sqrt{3}}$, $x = \frac{-9 - 4\sqrt{3}}{3}$ $y = \frac{9 + 16\sqrt{3}}{9}$

CAREFUL WITH CALCULATION DISPLAY: $-\frac{9 + 4\sqrt{3}}{3}$

∴ THE COORDINATES OF THE STATIONARY POINTS ARE

$\left(\frac{-9 + 4\sqrt{3}}{3}, \frac{9 - 16\sqrt{3}}{9}\right)$ AND $\left(\frac{-9 - 4\sqrt{3}}{3}, \frac{9 + 16\sqrt{3}}{9}\right)$

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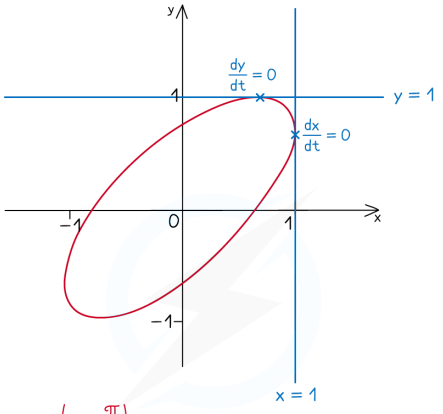
- You may also be asked about horizontal and vertical tangents
  - At horizontal (parallel to the **x**-axis) tangents, **dy/dt = 0**
  - At vertical (parallel to **y**-axis) tangents, **dx/dt = 0**



$x = \cos\left(2t - \frac{\pi}{4}\right)$

$y = \sin\left(2t + \frac{\pi}{2}\right)$

- $x = -1$ IS ALSO A VERTICAL TANGENT
- $y = -1$ IS ALSO A HORIZONTAL TANGENT

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